A child with cough

Dr. Ka-ka SIU

Dept of Paediatrics & Adolescent Medicine

Cough

- Under both voluntary and involuntary control
- Clears the airways of secretions
  - Primary mechanism for secretion removal when the respiratory cilia are damaged by inflammation
- Cough reflex protects the airway following inhalation of foreign material (food, secretions etc)
- Cough receptors located within the epithelium of the pharynx, larynx, trachea and at the major bronchi
  - Stimulated mechanically, chemically, thermally
  - Stimulated by local mediators: histamine, prostaglandins, leukotrienes and by local bronchoconstriction
- Reflex arc: afferent fibres from vagus nerve to cough centre in upper brain stem; efferent fibres from vagus nerve and spinal cord to larynx, diaphragm and abdominal muscles to produce a cough
- Higher cortical control of visceral reflex
  - Cough inhibition and voluntary cough
- Cough receptors also in the pharynx, paranasal sinuses, stomach and external auditory canal
  - Source of persistent cough may not be in the lungs

Causes of cough

- Respiratory
  - Upper respiratory tract infection
  - Postnasal drip syndrome (Upper airway cough syndrome)
  - Asthma
  - Aspiration
  - Pneumonia/pneumonitis
  - Bronchiectasis
- Non-respiratory
  - Heart failure
  - Gastroesophageal reflux
  - ACE inhibitor
  - Psychogenic/habit cough

A child with a cough comes to see you, what would you like to know?

Questions to ask

- Who
- What
- How
- When

Questions to ask

- **Who** is this child?
  - Age
  - Previously healthy?
  - Immunocompromised? HIV infected?
  - Neurologically impaired?
  - Atopic?
  - Living at home? Institution?

Questions to ask

- **What** is the cough like?
  - Dry?
  - Productive?
    - (Blood stained?), (colour?)
  - Barking? Brassy? Wheezy?
  - Paroxysmal?

Dry Vs. Productive

- Meaning of `productive'
- Beware of `sputum sound', what does that really mean?
- Mucus is produced by goblet cells and submucosal glands under normal situation and may increase in illness (infection, smoking etc.)
  - Nasal mucus
  - Airway mucus
    - Trachea, bronchus, bronchioles
    - `sputum' or `phlegm'
- Exudates (protein-rich fluid leaked from capillaries) in alveoli due to inflammatory response due to usually infection

Questions to ask

- **When** does the cough happen?
  - After feeding?
  - In the middle of the night?
  - When the weather changes and turns cold?
  - When someone smokes?
  - With exertion?
  - Associated with fever?
  - Associated with runny nose?
- **What** makes it better or worse?
- **How** long has the cough been?

How long has the cough been?

- The most common stimulus of cough
  - Irritation or inflammation of the respiratory epithelium
- Acute cough
  - A recent onset of cough lasting less than 3 weeks

Acute cough

- Acute infection
  - Viral upper respiratory tract infections account for most acute cough at all ages
  - Self-limiting
  - Usually lasts 1-3 weeks
  - URI: viral (influenza, respiratory syncytial virus, parainfluenza, adenovirus, rhinovirus, human coronavirus, human metapneumovirus, bocavirus)
  - LRI: viral, bacterial (*Streptococcus pneumoniae*, *Haemophilus influenzae*, Pseudomonas if immunocompromised, chlamydia if neonate)
- Exacerbation of pre-existing condition
  - Asthma, bronchiectasis, upper airway cough syndrome (postnasal drip)
- History and physical examination

Physical Exam of a patient with an acute cough

- Main issues to sort out:
  - Severity
    - Does this patient need supportive treatment?
  - Etiology
    - Infectious vs. noninfectious
    - If non-infectious:
      - allergy? (nasal? Airway? Both?)
    - If infectious: can you rule out lower respiratory tract infection?
      - Likely organism

Physical examination

- Temperature (Fever)
- Vital signs
- Respiratory distress?
  - Respiratory rate
  - Retraction/insucking/use of accessory muscles
  - Cyanosis, oxygen saturation
  - Dyspnea/Shortness of breath
- Chest exam: deformity, percussion, auscultation (wheeze, crepitations, rhonchi)
- Associated finding: skin rash, eczema, tonsils, lymph nodes, rhinorrhea

Acute cough

- The absence of fever, tachypnea and chest signs appears to be most useful for ruling out lower respiratory involvement
- Most children with cough due to a simple URI do not need any investigations
- A CXR should be considered in the presence of
  - lower respiratory tract signs (+/-)
  - relentlessly progressive cough (eg. past 2 week point)
  - hemoptysis or (an undiagnosed chronic respiratory disorder)
- +/- other investigations depending on your ddx
  - eg.,CBC and differentials
  - Nasopharyngeal aspirates for common viruses and mycoplasma
  - sputum for gram stain and culture if old enough
  - Blood culture has low detection rate even if bacterial etiology

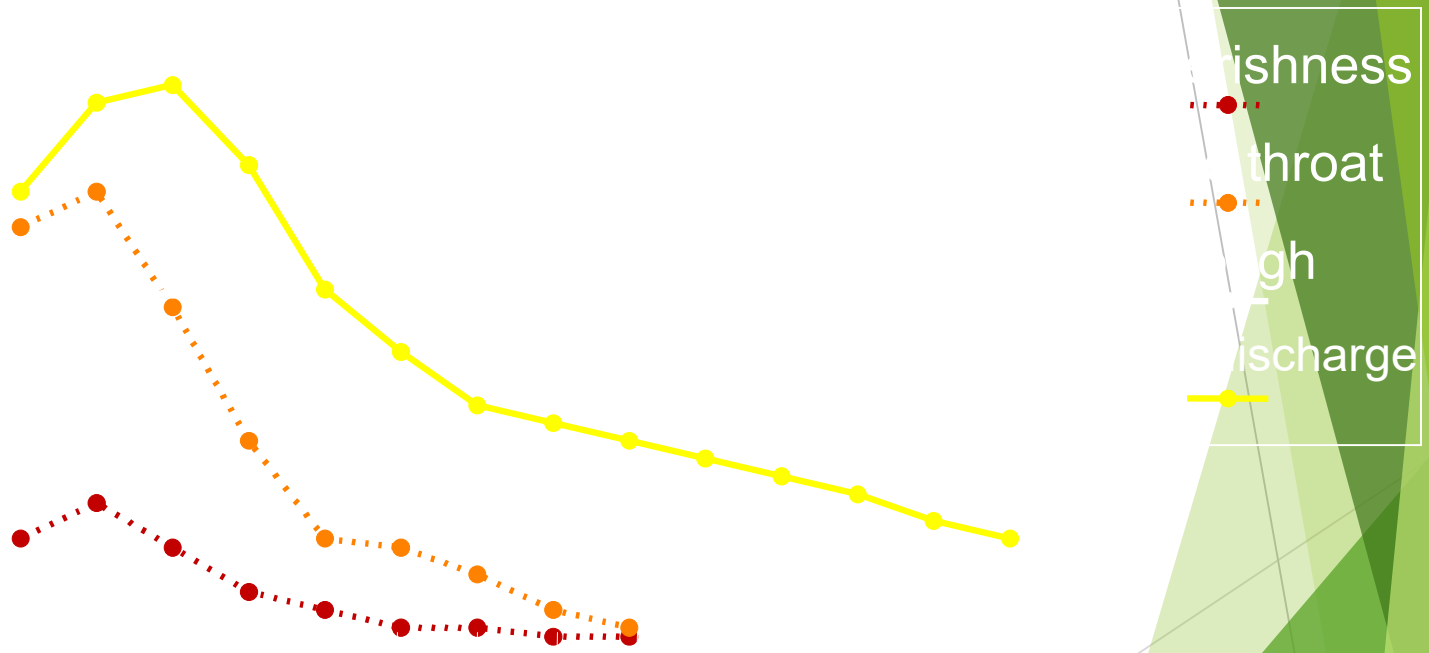
Approach to try to arrive at a specific diagnosis for acute cough

Question	Features	Likely common dx
Is this an acute URI?	Coryzal symptoms Fever sorethroat	URI
Is this a croup syndrome?	Stridor, `barking' or `croupy cough', hoarseness, +/- fever	Viral croup Recurrent spasmodic croup Bacterial tracheitis
Is this a lower respiratory tract illness?	Tachypnea (> 60 for <2m, >50 for 2-12m, >40 for >1 yr), respiratory distress with increased work of breathing, chest signs (crepitations or wheeze/rhonchi), fever	Acute bronchiolitis Pneumonia (viral, bacterial) Asthma
Is this an allergic/atopic illness?	Seasonal and diurnal variation, association with rhinitis, posture, `clearing of throat', triggers (dust, pollutant, pollen etc.)	Post-nasal drip from allergic rhinitis Reactive airway/asthma
Is this an acute exacerbation of a chronic respiratory disorder?	Failure to thrive, finger clubbing, chest deformity, features of atopy	To be continued….

Chronic cough

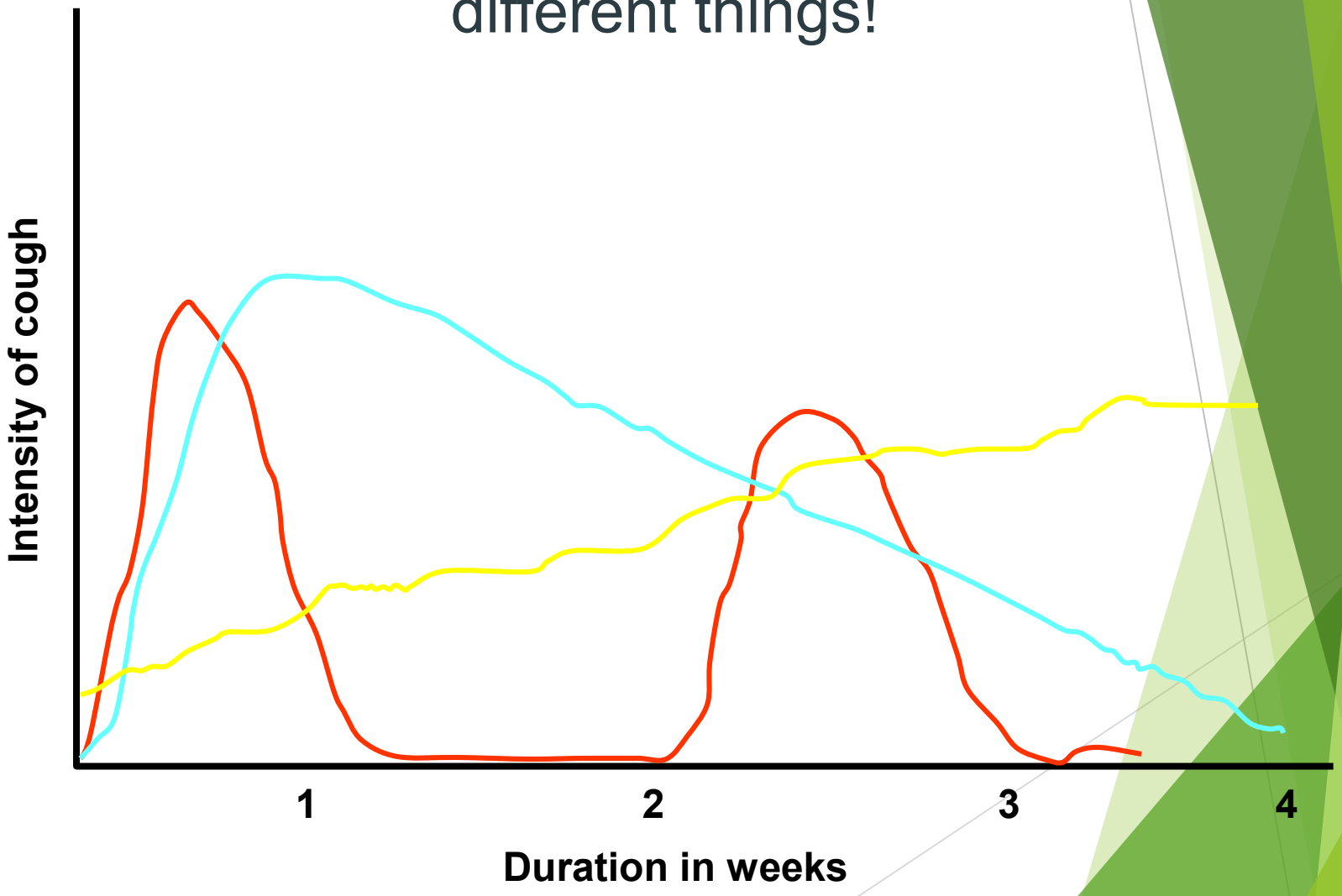
- American College of Chest Physician Guideline (Chest 2006)
- Pediatric (<15yrs) chronic cough: daily cough lasting for > 4 weeks
  - No studies that clearly define
  - Time frame chosen due to natural history of URI in children (lasting 1-3 wks in children)
    - 1 study: 10% preschool children still coughing 25d after URI
  - Different from definition in adults
- British Thoracic Society (Thorax 2007)
  - A cough lasting > 8 weeks
  - Recognizes a `grey' area of `subacute cough' between 2-8 weeks
- Subdivided into specific cough (cough with S&S suggestive of associated problem) and nonspecific cough (dry cough in absence of an identifiable respiratory disease of known etiology)

Duration of Symptoms in 139 Rhinovirus Colds in adult volunteers



Gwaltney, JAMA 1967;202:158

Duration and intensity
Be specific---`same' duration may mean different things!



Question	Examples	DDX
How did the cough start?	Very acute or after choking	Inhaled Foreign body
	With URI	Post-viral cough
When did the cough start?	Neonatal onset	Aspiration, congenital malformation, Cystic fibrosis, primary cilial dyskinesia
What is the quality of the cough?	Productive (moist/wet)	Chronic suppurative disease (eg. Bronchiectasis)
	Paroxysmal spasmodic cough w/wo `whoop'	Pertussis
	Hemoptysis	TB in adolescents, bronchiectasis, AV malformation
	Bizzare honking cough which increases with attention	Psychogenic cough
	dry repetitive, disappears with sleep	Habit cough
What triggers the cough?	Exercise, cold air, early morning	asthma
	Feeding	Recurrent aspiration

Specific cough--Associated with signs and symptoms of an underlying respiratory or systemic disease

Abnormality	Examples of etiology
Wheeze	Intrathoracic airway lesion (eg., asthma, foreign body)
Crepitation	Parenchymal disease
Chest pain	Arrhythmia, asthma, increased respiratory distress (parenchymal disease)
Chest wall deformity	Chronic airway or parenchymal disease
Digital clubbing	Chronic suppurative lung disease
Daily moist/productive cough	Suppurative lung disease
Failure to thrive	Serious systemic including pulmonary illness
Feeding difficulties	Serious systemic including pulmonary illness, aspiration
Hypoxia/cyanosis	Airway or parenchymal disease, cardiac disease
Neurodevelopmental abnormality	Aspiration lung disease
Recurrent pneumonia	Immunodeficiency, congenital lung abnormalities, tracheo-esophageal H fistula

Specific cough---Typical cough association

Whooping cough	Pertussis
Barking or brassy cough	Croup, tracheomalacia, habit cough
Paroxysmal (with or without whoop)	Pertussis, mycoplasma, parapertussis, virus
Nocturnal	Upper or lower respiratory allergy, sinusitis
Tight (wheezy)	Reactive airway
Staccato	Chlamydia
Honking	Psychogenic/habit

Specificity and sensitivity only validated in brassy vs. nonbrassy cough for tracheomalacia (0.57 and 0.81)

Clues some questions provide

- Who is this child?
  - Age
  - Previously healthy?
  - Immunocompromised?
  - Neurologically impaired?
  - Atopic?
  - Living at home? Institution?
- What is the cough like?
  - Dry?
  - Productive?
  - Blood stained?
  - Barking? Brassy?
  - Wheezy?
- How long has the cough been?
- When does the cough happen?
  - After feeding?
  - In the middle of the night?
  - When the weather turns cold?
  - With exertion?

likely organisms differ, types of problem differ
Community-acquired vs. hospital-acquired organism
Opportunistic organism
Aspiration pneumonia from swallowing incoordination
Asthma or postnasal drip
Community-acquired

Mycoplasma, pyschogenic, asthma
Pneumonia, bronchiectasis
Pneumonia, bronchiectasis, TB, excoriated airway
Croup
Asthma/acute bronchiolitis
Acute vs. subacute vs. chronic

GER
Asthma, sinusitis
Asthma
Heart failure, asthma
Postnasal drip
Infection
Psychogenic cough/habitual cough

Clues from age of onset of cough

Infancy	Early childhood	Late childhood/adolescence
Aspiration	Aspiration	Aspiration (degenerative neuromuscular disease)
Reactive airway	Asthma	Asthma
Congenital malformation: laryngotracheomalacia or bronchomalacia, vascular compression (ring or sling innominate artery)	Bronchiectasis (Immunodeficiency, cystic fibrosis, postinfectious)	Bronchiectasis (Immunodeficiency, cystic fibrosis, postinfectious)
Infection: Chlamydia, pertussis, Tuberculosis, post- respiratory syncytial virus	Infection: Viral, tuberculosis, mycoplasma, fungal	Infection: Viral, tuberculosis, mycoplasma, fungal
Others: passive smoking, congenital heart disease	Sinusitis	Sinusitis
		Smoking (active and passive)
		Psychogenic
		Mediastinal Tumor

Non-specific cough

- Usually post viral cough or another episode of acute infection
  - Others: foreign body, asthma, GERD etc.
- Evaluate
  - for tobacco smoke and other pollutants
    - Prevalence of chronic cough in children < 11yrs with 2 smoking parents is 50%
  - Child's activity
  - Parental expectations and concerns
- Watch, wait and review: likely resolve without specific treatment
- Absence of specific cough pointers

Physical examination of a child with chronic cough

- Growth and development
- Respiratory distress
- Digital clubbing
- Upper respiratory tract
  - Signs of sinusitis (facial tenderness, persistent purulent nasal discharge)
  - Signs of allergic rhinitis (nasal obstruction, nasal discharge)
- Lower respiratory tract
  - Chest deformity (Harrison sulcus, pectus carinatum, barrel chest)
  - Hyperresonance
  - Crepitations, wheeze, rhonchi
- Cardiac: murmur or heart failure
- Skin: eczema

Investigations

- According to History and PE
- Developmental appropriate
  - CXR
  - Peak flow +/- lung function study
  - CBC with differentials
  - Mantoux test (MT5)/Purified Protein Derivative Test (PPD) for TB
  - Inteferon-based test for TB
  - Sputum or gastric aspirate for TB
  - HRCT
  - Nasal NO/ Cilia study / Sweat test
  - Immunoglobulin pattern
  - pH probe
  - Video fluoroscopy
  - Bronchoscopy with BAL

Mantoux Test

1

Use a 1mL syringe to aspirate out **0.1 mL** of PPD RT 23.

2

Inject the PPD intradermally on the volar surface of the forearm. Position the syringe at a 10-15° to the forearm and insert just below the epidermis (about 2 mm).

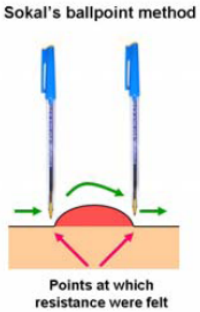
3

Remove the needle quickly. Do not massage or use dressing. A well-defined bleb of 6-10mm in diameter should be formed if injected correctly. If the bleb is <6mm, repeat the process 2.5cm from the first site.

4

Mark down the **site, date and time of injection**, both on the forearm and in patient's record.

5



After 48 to 72 hours, read the test result by marking down the transverse diameter of **induration**, not erythema, by Sokal's ballpoint method.

6

Measure the largest transverse diameter of induration and note down in millimeters (mm). **Do not just write test result positive or negative.**

Treatment

- Cough is a symptom telling you something is wrong
  - Find the cause
- Treat the underlying cause if indicated
  - Very diverse treatment
    - Bronchodilator, surgery, antibiotics, anti-TB, anti-fungal, stop smoking etc.

Viral infection

- No specific antiviral drug available in most viral infection
  - `bronchitis'
  - Pneumonia
- Anti-virals available for influenza
  - Amantadine, oseltamivir, zanamivir
  - Not to treat cough per se (neuraminidase inhibitors shown to shorten fever by 1.5 days)
  - Problem with increasing resistance

Bacterial infection

- Community acquired pneumonia
  - *Streptococcus pneumoniae, H. influenzae, Moraxella catarrhalis*
    - Augmentin
  - atypical': *Mycoplasma pneumoniae*
    - clarithromycin, doxycycline/quinolone
- Pneumonia in children with underlying disease
  - Eg. Pseudomonas, *staph aureus*

Should cough suppressants be used?

- A natural, protective, beneficial mechanism
- Critical review of clinical trials 1950-1991 (JAMA 1993)
  - Many antitussive studies performed on animals or on humans with an artificial cough (produced by citric acid or ammonia inhalation or chronic respiratory disease)
  - No good evidence to demonstrate effectiveness of over-the-counter cold medication in preschool children
  - (certain antitussive may work for adolescents and adults)
- Cochrane 2004 (7 pediatric studies, 516 children)
  - Studies in children with acute cough
    - Antitussive no more effective than placebo
    - No study on expectorants met inclusion criteria
    - 1 trial favoured mucolytic (Letosteine) over placebo from d4-d10
    - 2 studies showed no difference between antihistamine-decongestant combinations and placebo
    - 1 trial showed antihistamine was no more effective than placebo
  - Conclusion: no good evidence for or against the effectiveness of OTC medicine in acute cough

Examples of common ingredients in cough and cold medications

Antihistamine	Diphenhydramine, cyproheptadine, brompheniramine, Chlorpheniramine
Antipyretic and analgesic	Paracetamol, ibuprofen
Antitussive	Codeine, dextromethorphan, hydrocodone
Expectorant	Guaifenesin
Nasal decongestant	Ephedrine, phenylephrine, pseudoephedrine, phenylpropanolamine

Are they benign?

- Codeine and dextromethorphan are 2 antitussive agents frequently used in pediatrics
  - Work by acting centrally suppressing the medullary cough centre
- Codeine:
  - narcotic with addictive potential
  - Dose-related toxicity: respiratory depression and narcosis
  - Somnolence, ataxia, miosis, vomiting, rash, swelling, itching
- Dextromethorphan:
  - considered nonaddictive but abused by teenagers---ingesting huge doses results in bizarre behaviour
  - CNS depression
- Combination commonly used:
  - Nasal decongestants, antihistamine, cough suppressants, expectorants
  - No FDA approved dosing exist for children < 2 years of age
  - Death and morbidity reported over and over again worldwide including HK

- In 1997 the American Academy of Pediatrics issued recommendations:
  - There are no well controlled scientific studies to support efficacy of codeine-containing and dextromethorphan-containing antitussives in children and indications use have not been established
  - Suppression of cough in many pulmonary diseases may be hazardous and contraindicated. Cough due to acute viral infection is short-lived and can be treated with fluids and humidity
  - Dosages for children are extrapolated from adult data. Adverse effects and overdosage in children are reported.
  - Education of parents about lack of proven effects and potential risks is needed
- In 2006, the American College of Chest Physicians advised doctors to refrain from recommending cough suppressants and other `over-the-counter' cough medications for young children

11 Oct 2007: Manufacturers voluntarily withdrew OTC cough cold medicines for kids < 2 yrs from the market in the US

Ingredients and Marketing of Some of the Available Pediatric Cough and Cold Products.*

Ingredients	Brand Name	Marketed Use
Antihistamine	Pediacare	Nighttime cough
	Triaminic	Cough and runny nose
Antitussive	Robitussin	Long-acting cough
Decongestant	Dimetapp	Decongestant
Antihistamine and antitussive	Robitussin	Long-acting cough and cold
	Tylenol Plus	Cough and runny nose
Antihistamine and decongestant	Dimetapp	Cold and allergy
	Pediacare	Nighttime multisymptom cold
	Triaminic	Nighttime cough and cold
Antitussive and decongestant	Dimetapp	Decongestant plus cough
	Pediacare	Multisymptom cold
	Triaminic	Daytime cough and cold
Decongestant and expectorant	Dimetapp	Cold and chest congestion
	Triaminic	Chest and nasal congestion
Antihistamine, antitussive, and decongestant	Dimetapp	Cold and chest congestion
	Tylenol Plus	Flu
	Tylenol Plus	Multisymptom cold
Antitussive, decongestant, and expectorant	Robitussin	Cough and cold

* The antihistamines include brompheniramine, chlorpheniramine, and diphenhydramine; the antitussive is dextromethorphan, the decongestant is phenylephrine, and the expectorant is guaifensin. All formulas of Tylenol Plus also contain acetaminophen.

The responsibility of the doctor

- A pediatrician is a child's advocate
  - We treat the child, not the parents
  - Have to work with the caretakers
- Our role to educate
  - Natural course of the illness
  - Preventive measures when applicable
  - Symptomatic relief is not everything
  - Appropriate therapy includes avoiding inapproprate therapy
- Patients are not (only) customers

Summary

- Causes of cough in children may be different from adults
- Manifestations may be different from adults
- A child with a cough may have a self-limiting problem or a serious illness
- Importance of history and physical examination
- All children with chronic cough should have thorough clinical review to identify possible underlying respiratory and/or systemic illness
- Cough treatment should be based on etiology
- No evidence to support medicine for symptomatic relief
- Importance of education